Venkata Lakshmi Parimala Pasupuleti

Senior Fullstack Engineer · Agent Observability · Python · FastAPI · Distributed Systems · LLM Evaluation · React · AWS

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SUMMARY

Senior Software Engineer with 8+ years building production backend services, distributed systems, and data pipelines - and two years focused on production agentic AI. Built **AgentProbe**, an open-source agent evaluation and observability toolkit: 5-metric eval engine (correctness, hallucination rate, retrieval quality, completeness, compliance), LLM-as-judge scoring, CI-integrated regression harness with golden datasets, and trace-level monitoring across all agent runs - exactly the kind of agent trust infrastructure Monte Carlo provides to enterprises. Built **Leasel**, an 8-agent LangGraph production system with trace-level observability, structured audit logging, and a CI regression gate that catches degradations before they reach users. At Entrata (8 years, 100K+ users), owned multi-tenant SaaS backend services, a RabbitMQ-based distributed async processing system, data pipelines transforming federal datasets at scale, and third-party API integrations in a regulated production environment. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Agent Observability | LLM Evaluation & Monitoring | Python · FastAPI | Distributed Systems | Data Pipelines | Production Agentic AI | LangGraph · LangChain | LLM-as-Judge | CI/CD · Regression Harness | AWS · S3 · EC2 · Lambda | React · TypeScript | PostgreSQL · SQLite · Redis | RabbitMQ · Kafka | REST APIs · OpenAPI | Docker · Kubernetes | Multi-Tenant Architecture | ML Platform | Java · PHP · Node.js

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, data analysis pipelines, and API integration layers connecting AI components to application-facing backend and reporting systems.
- Applying debugging and troubleshooting practices across the full stack to improve reliability of data-driven and AI-powered features.

AI/ML Engineer Intern

May 2025 - Aug 2025

Magizhs Tech Solution

- Built an agentic mission-planning system (LangGraph StateGraph: parse → plan → retrieve → validate → conditional routing) with hard validation guardrails enforcing constraints before export - non-compliant routes surface grounded rejection reasons; deployed MCP server connecting agentic outputs to downstream enterprise tooling.
- Implemented a RAG layer over FAA Part 107 regulations with cited-source answers; engineered the full ingestion, chunking, retrieval, and FastAPI serving pipeline from corpus to production endpoint.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ users, multi-tenant)

- **Built and operated a RabbitMQ-based distributed async processing system** for gross-rent updates across thousands of tenant records - designed retry logic and idempotency guarantees eliminating double-processing; owned through production incidents and annual rent-schedule changes.
- **Owned the annual HUD income-limit data pipeline for 7 years** - few thousand federal rows → ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units; designed baseline validation, reconciliation, rollback preparation, and post-release monitoring every cycle.
- **Optimized 200+ SQL queries (30% performance gain)** through execution plan analysis, indexing, partitioning, and stored procedure refactoring across complex multi-tenant production schemas for eligibility, financial, and document workflows.
- **Led Affordable Housing module integration into the AWS S3 document migration** - defined input contracts, validated metadata correctness post-cutover; 100% PHI compliance via Libsodium-encrypted pipelines, zero security incidents across the full resident data migration.
- Shipped 20+ full-stack production modules (Python, PHP, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; led microservices redesign using Docker - 50% scalability improvement, 30% deployment time reduction; built internal debugging tools reducing debugging time by 40%.
- Mentored 5+ engineers; conducted 300+ peer code reviews; maintained bug count below 10 for 6+ consecutive months.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built property management backend modules (PHP, Python, Java, PostgreSQL) including income-limits import (1.6M HUD/Tax Credit records), document management, and compliance workflows.
- Led cloud migration from legacy storage to AWS S3; redesigned document storage architecture migrating hundreds of thousands of records to a modern schema with 6 months of post-deployment stabilization.

PROJECTS

AgentProbe - Open-Source Agent Evaluation & Observability Toolkit

2025 | github.com/venkatalakshmiparimala/AgentProbe

- **Built the core of what Monte Carlo's agent trust platform does**: a 5-metric eval engine (factual accuracy, hallucination rate, retrieval quality, completeness, compliance), LLM-as-judge scoring with configurable rubrics, CI-integrated regression harness with golden datasets, and offline + online evaluation loops - monitors and troubleshoots AI agent behavior in production.
- 15-endpoint FastAPI backend; dynamic LLM routing across OpenAI, Anthropic/Claude, Azure OpenAI, Groq, and Ollama; fully local, zero vendor lock-in; used to evaluate Leasel across 1,240+ compliance test cases with structured failure analysis.

Leasel - Production Agentic AI Platform

2026 | Python, LangGraph (8-agent), FastAPI, SQLite, CI/CD

- **8-agent LangGraph production system** (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) with durable state, conditional routing, trace-level observability, structured audit logging, and HITL escalation when confidence is insufficient.
- CI regression gate with LLM-as-judge scoring catches agent degradations before they reach users; 93.6% precision and 87.9% F1 across 1,240 compliance test cases (10 rules, 5 US states + federal).

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017